

Optimal Interconnection Planning of Community Microgrids With Renewable Energy Sources

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Abstract—The optimal planning of the interconnected network of multimicrogrids is discussed in this paper. The interconnection planning will enhance the reliability and the economic operation of a community of microgrids. The proposed approach will apply a probabilistic minimal cut-set-based iterative methodology for the optimal planning of interconnection among microgrids with variable renewable energy sources. The optimal planning takes into account various factors including the economics, reliability, and variability of renewables, network- and resource-based uncertainties, and adaptability to accommodate the prevailing operating concerns. A clustering-based method is considered for analyzing the variable data concerning the potential deployment of renewable energy in microgrids. The proposed interconnection planning methodology is applied to a six-microgrid system and the planning results are discussed. The numerical results demonstrate that the proposed interconnection planning methodology will determine an optimal topology accurately and efficiently for a cluster of microgrids, and show that the proposed adaptive planning methodology can easily be applied to practical microgrid applications.

Index Terms—Community microgrid, distribution network planning, microgrid planning, minimum cut-set, reliability.

NOMENCLATURE

Variables

$C_{\text{AnnPay}}, C_{\text{Interrupt}}$	Annualized project capital payment and annual customer energy interruption cost.
$\mathbf{R}_i$	Reliability indices matrix from iteration- i .

Indices and Set

i	Index of iteration in the proposed method.
s	Index of wind state ($s = 1, \dots, N_s$).
t	Index of time period ($t = 1, \dots, N_t$).

y	Index of year ($y = 1, \dots, N_y$).
T	A set of time periods $\{t\}$ where in each t both deficient and surplus microgrids exist.

I. INTRODUCTION

MICROGRIDS are self-controlled entities which facilitate the penetration of renewable energy and distributed energy resources (DERs) for economic and reliability purposes [1]–[3]. Microgrids, which are operated in either grid-connected or island mode, are vital components of smart power distribution networks for managing the power system resilience in extreme conditions. A cluster of interconnected microgrids, which are tied through physical connections, represents a community of microgrid for enhancing reliability, resilience, and economics of power systems [4], [5].

In [5], we proposed an optimal coordination strategy for regulating power exchanges among paralleled microgrids in a community microgrid. The proposed strategy initiates power exchanges among microgrids as system conditions change, and corrects frequency and voltage deviations in participating microgrids, while maintaining an economically viable optimal microgrid operation. In [5], we developed a coordinated strategy with an underlying assumption that all participating microgrids were interconnected. In practice, the interconnection topology ought to be properly planned for enhancing the economic, resilience, and reliability signals in multimicrogrid operations. This topic will be investigated in this paper.

In power system planning, most of the transmission interconnection planning studies consider a least-cost solution [6]–[8]; while other studies propose a comprehensive review of interconnection alternatives [9]. Khodaei and Shahidehpour [10] studied a transmission expansion problem that considers the integration of microgrids. Compared to transmission planning studies, the extent of research articles on microgrid topology is relatively limited. A preliminary study of the microgrid network topology which uses integer programming is presented in [11]. The reliability-based microgrid design, which focuses on the reconfiguration of a single microgrid is proposed in [12].

In this paper, we investigate the optimal planning of network interconnection for multimicrogrids. Additional challenges are imposed in this paper by considering the distinctive features of distributed microgrids, i.e., each microgrid is considered an autonomous entity that combines generations and load

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controls for satisfying local objectives. The optimal planning option may not always be represented by a least-cost solution. Instead, as motivated by [9], we take a comprehensive consideration of prevailing factors including the economics, reliability, and variability of renewables, network- and resource-based uncertainties, and the adaptability to accommodate various operating concerns.

Major contributions of this paper include: 1) a new research direction for the optimal planning and the operation of multimicrogrids; 2) an optimal planning of microgrid interconnection which considers a clustering method for representing the deployment of variable renewable energy; and 3) a reliability-based solution of interconnection system which utilizes the minimal cut-set (MCS) method for the optimal planning of probabilistic microgrids.

The rest of this paper is arranged as follows. Section II highlights the benefits of the interconnection. To obtain these benefits, the proposed planning methodology is discussed in Section III and demonstrated by case studies in Section IV. Finally, Section V concludes this paper.

II. INTERCONNECTION OF MULTIMICROGRID

There are numerous reasons as to why microgrids ought to be interconnected. The first and primary benefit of microgrid interconnection is to share reserves in critical conditions (e.g., loss of a major generation) for lowering the chance of system collapse, minimizing emergency load shedding requirements, and enhancing the overall system reliability.

The second benefit of interconnection is the economic dispatch in multimicrogrid systems which can occur in either grid-connected or island mode. For example, a microgrid with excess solar power generation can assist its neighboring deficient microgrid, which would benefit both microgrids economically.

The third benefit is sharing storage and ancillary functions. For instance, large energy storage systems in a microgrid can serve critical loads in several interconnected microgrids before thermal generation units in individual microgrids can pick up the respective loads. As a functional microgrid, the microgrid at Illinois Institute of Technology (IIT) is equipped with an 8 MW natural-gas turbine unit (with black-start capability) and a 500 kWh battery storage. So when interconnected to another microgrid, the IIT microgrid can provide black-start function to both entities while accessing other types of local resources which can benefit multimicrogrids.

The fourth benefit concerns the microgrid resilience. Microgrids may encounter extreme challenges in emergencies including unintentional islanding due to a major disturbance at the utility side. Adequate and strategic interconnection of microgrids can mitigate the impact of such emergencies.

Furthermore, a large microgrid can be partitioned into interconnected nanogrids (loops) for control purposes. For example, a distributed generation system in a microgrid may be partitioned into several nanogrids for enhancing the local reliability [12].

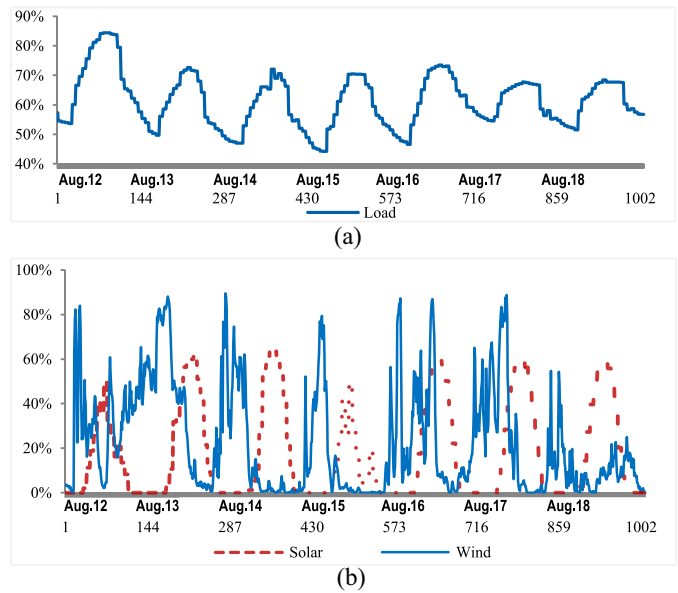


Fig. 1. Time-series in a typical week in Aug 2013 for (a) load and (b) solar and wind.

III. INTERCONNECTION PLANNING METHODOLOGY

In this section, we propose our interconnection planning methodology that supports the various technical features pertaining to the optimal operation of multimicrogrids.

A. Data Processing

Renewable energy like solar and wind exhibits inherent uncertainty and variability, which would affect the overall power system planning. Such impacts become more prevalent with the integration of renewable generation and distributed generation systems in multimicrogrids. To address the impact of uncertainty, which results from the variability of such resources, we discuss the data processing in this section as the first step in developing our proposed methodology.

1) *Generation and Load Time-Series*: Fig. 1 shows the load, solar, and wind energy time-series taken in 10-min intervals at the IIT microgrid. The data are presented as a percentage of respective annual peaks. Here, the hourly load exhibits a more distinctive correlation with solar and a weaker correlation with wind energy. The correlation coefficients are stated as: $\text{Cov}(\text{load}, \text{solar}) = 0.355$ and $\text{Cov}(\text{load}, \text{wind}) = 0.049$, which can verify the observation.

To further investigate the different characteristics of solar and wind generations, we apply clustering analysis [13], [14] to partition the original data into mutually exclusive clusters in which the sum of squared Euclidean distances between each point and its nearest centroid is minimized. The cluster centroids are used to investigate the characteristics of wind and solar energy in microgrids. Here, the data given in Fig. 1(a) are converted to 365 daily time-series, then partitioned using the k -means clustering method into 12 clusters with the centroids shown in Fig. 2.

In Fig. 2(a), the clustered solar curves are smooth which adopt a similar profile. The solar curves shift along x -axis

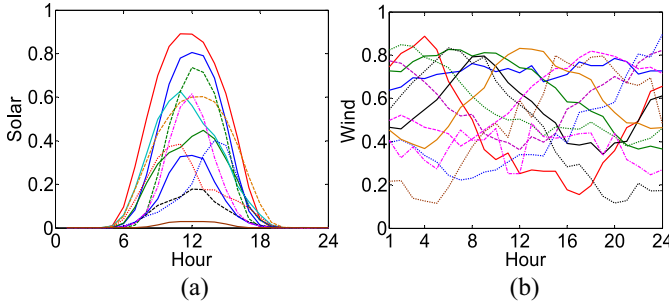


Fig. 2. Clustering daily time-series for (a) solar and (b) wind energy.

which represents seasonal variations of sunrise and sunset. As a comparison, the clustered wind curves in Fig. 2(b) exhibit a more unpredictable pattern. Here, the difference between Fig. 2(a) and (b) is mainly due to the fact that solar irradiation adopts a 24-h cycle while wind energy is much more likely to be affected by short-term and random weather factors. Similar conclusion in [15] demonstrates a significant correlation between temporal solar and load profiles; and in order to preserve the correlation, solar and load variations are modeled as deterministic time-series using historical data over a given period.

Based on our clustering analyses, we focus our attention further on the variation of wind energy; while solar energy and load data are processed as synchronized deterministic time-series in which the data for each time period is expressed by

$$\bar{x}_t = \frac{1}{N_y} \sum_{y=1}^{N_y} x_{t,y}, (\forall t = 1, \dots, N_t). \quad (1)$$

Considering the periodic characteristic of solar energy and its high correlation with the system load, the proposed clustering approach will reasonably reflect the variation of solar and load time-series and adds more realism to our microgrid analyses by offering a simple way of preserving the solar-load correlation in the proposed planning methodology. Note that the proposed methodology could also consider other uncertainties pertaining to energy sources and load, which will be discussed in Section III-B.

2) *Simplified Probabilistic Model for Wind Energy*: Several methods were proposed in reliability studies to characterize wind energy variations. These methods are categorized into: a) probability distribution function based method [16]; b) Monte Carlo simulation; and c) simplified wind model (SWM) [17]. The SWM represents the wind speed distribution by simplified probabilistic states based on a normal distribution, which is sufficiently accurate for the microgrid planning studies. The probabilistic SWM is shown in Fig. 3, where the midpoint in each state (W_s) is defined in Table I with a probability that is calculated from historical or simulated data. If such data are unavailable, one can use the probability values given in [17]. Using SWM, only the annual mean and standard deviation (μ and σ) of on-site wind speed are required. To obtain the wind energy, the modeled wind speed is applied to the wind turbine power curve

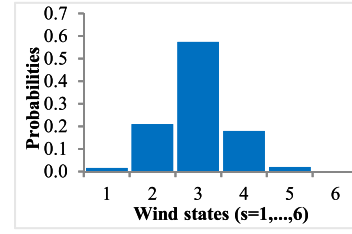


Fig. 3. Probabilistic SWM.

TABLE I
MIDPOINT VALUE FOR EACH WIND STATE IN FIG. 3

State-s	W_s
1	$\mu + 2 \cdot (5\sigma / 3)$
2	$\mu + (5\sigma / 3)$
3	μ
4	$\mu - (5\sigma / 3)$
5	$\mu - 2 \cdot (5\sigma / 3)$
6	$\mu - 3 \cdot (5\sigma / 3)$

TABLE II
DEFINITION OF MICROGRID STATUS

Status	Code	Without load shedding	With load shedding
Surplus	1	$P_t \geq L_t$	$P_t \geq L_t - \Delta L_t$
Deficient	0	$P_t < L_t$	$P_t < L_t - \Delta L_t$

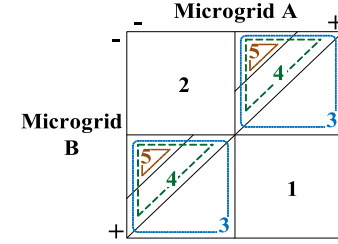


Fig. 4. Venn diagram for multimicrogrids.

expressed by (2), where parameters A , B , and C are defined in [18]. The turbine has a nonzero power output only when wind speed (w) is between cut-in and cut-out values (V_{ci} and V_{co}), and the output saturates above the rated speed (V_r)

$$P = \begin{cases} P_{\text{rated}} \cdot (A + B \cdot w + C \cdot w^2) & V_{ci} < w < V_r \\ P_{\text{rated}} & V_r < w < V_{co}. \end{cases} \quad (2)$$

The SWM approach is applied for the interconnection planning in this paper in which the number of SWMs equals the number of time periods (N_t). We only adopt states 2–4 because states 1, 5, and 6 in Fig. 3 have small probabilities. For each time period, the values of μ and σ are calculated using historical wind speed data and then SWM is established.

B. Proposed Probabilistic MCS Method

Table II defines the status of a microgrid in a multimicrogrid system, where P_t , L_t , and ΔL_t are the total generation, load and load shedding in t . Here, 1 and 0 denote the surplus and deficient status, respectively.

Bases on the definition given in Table II, the reserve margins in two autonomous microgrids (A and B) are studied using a Venn diagram [19] shown in Fig. 4, in which we do not consider the constrained interconnection capacity as most

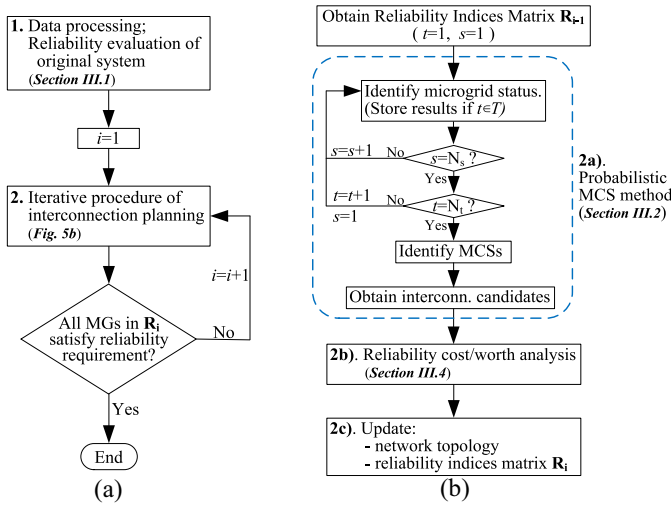


Fig. 5. Proposed planning methodology. (a) Overall structure. (b) Detailed information for step 2.

microgrid interconnections would be practically large. The areas in Fig. 4 represent possible reliability cases as follows.

- 1) Both microgrids are surplus (positive reserve margin).
- 2) Both microgrids are deficient (negative reserve margin).
- 3) One microgrid is deficient; the other is surplus and able to provide generation assistance.
- 4) Subset of (3) where the generation assistance is offered at the expense of load shedding in the surplus microgrid.
- 5) Subset of (4) where the generation assistance is limited by the allowable load shedding.

The MCS-based methods were used before for multi-area generation system reliability studies and transmission expansion planning [19]. In this paper, we propose a probabilistic MCS method which is illustrated as step 2a) in Fig. 5(b). In this method, the processed generation and load data are used to calculate the reserve margin of each microgrid in each wind state and time period. Then a probabilistic combination of microgrid status in each time period is obtained based on the definition introduced in this section. Here, the status and associated probabilities are stored for time period t , where $t \in T$ and T is a set of time periods in which both deficient and surplus microgrids exist. Next, probabilistic MCSs are identified using the microgrid status. In this probabilistic MCS method, a possible interconnection between two microgrids represents a component which, if not planned, represents the component failure; and load shedding in a microgrid represents the system failure. Each MCS is assigned a probability $\text{Prob}(s, t)$, expressed by (3), in which $\text{Prob}(s)$ is the wind-state probability derived from SWM and $\text{Prob}(t)$ is the probability of time period t . By assigning probabilities to MCS values, the renewable generation variability is incorporated into the interconnection planning process. The proposed reliability-based planning method can similarly consider the uncertainties of other energy sources or loads by multiplying the associated probabilities given

$$\begin{cases} \text{Prob}(s, t) = \text{Prob}(s) * \text{Prob}(t) \\ \sum_s \text{Prob}(s, t) = 1, \quad \forall t = 1, \dots, N_t. \end{cases} \quad (3)$$

In order to create interconnection candidates, the accrued MCS probability is assigned to each line included in that MCS.

So, a list of candidate lines is compiled in which the lines are ranked by their respective acquired probabilities. The rationale for ranking these lines by their total acquired probabilities lies on the following conjecture: the higher the probability of k th line, the higher the chances that the k th line will contribute to both the reliability of deficient microgrids and the economic operation of surplus microgrids. Finally, the top-ranked lines are selected as the interconnection planning alternatives, which is the output of step 2a) in Fig. 5(b).

C. Structure of the Proposed Planning Methodology

Fig. 5(a) illustrates the overall proposed planning methodology, which features an iterative procedure. Each iteration aims to improve the reliability of a microgrid with the worst reliability performance considered within the community. The number of iterations in this proposal is not essentially correlated with the number of microgrids that initially fail the reliability evaluation; rather, the iteration terminates when all microgrids satisfy reliability requirements. Multimicrogrids can change from reliability-unsatisfied to reliability-satisfied in each iteration. Step 2 (iterative planning) is elaborated in Fig. 5(b).

In Fig. 5(b), step 2a) is the proposed probabilistic MCS method discussed in Section III-B. The output of step 2a) is a list of selected interconnection options (ranked by their total acquired probabilities). In step 2b), the reliability cost/worth of these options is analyzed, as discussed in Section III-D, for determining the optimum solution in each iteration. Finally, using the current interconnection structure, the network topology and reliability indices are updated at the end of each iteration.

D. Reliability Cost/Worth Analysis

The reliability cost/worth analysis [9] balances the benefits of system improvements with the incurred investment costs, and offers system planners the additional opportunities for examining a variety of valuable information from different viewing angles. The analysis contributes to the system adaptability by accommodating various operating concerns that would lead to rational decision-making in an uncertain environment.

The reliability cost/worth analysis shown in Fig. 6(a) is an efficient method for the power system planning. The reliability cost is annualized as $C_{\text{AnnPay}} = \text{Project_investment} \times \text{CRF}$, where capital recovery factor $\text{CRF} = r(1+r)^n / [(1+r)^n - 1]$ and r is yearly discount rate. The reliability worth in Fig. 6(a) is derived as the annual customer interruption penalty ($C_{\text{Interrupt}}$) that will be lowered by a higher reliability performance.

We modified the reliability cost/worth analysis alternatives, which is the output of stepcurve to the one in Fig. 6(b), in which the reliability worth is interpreted as the annual incremental saving (ΔC) determined by

$$\Delta C = \Delta C_{\text{Interrupt}} + \Delta C_E \quad (4)$$

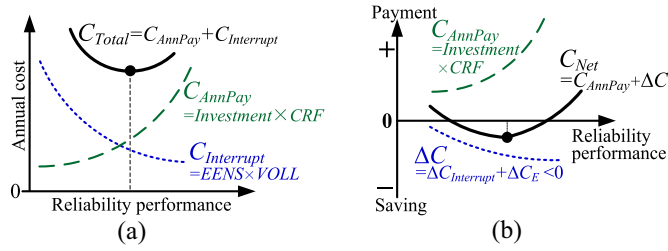


Fig. 6. Reliability cost/worth analysis. (a) Standard. (b) Modified.

where $\Delta C_{Interrupt}$ and ΔC_E are the annual incremental customer interruption penalty and the annual incremental energy cost considering renewables, respectively. A negative $\Delta C_{Interrupt}$ indicates that the customer interruption penalty is reduced by the interconnecting of multimicrogrids. Furthermore, a negative ΔC_E denotes that the total energy cost is reduced by a more efficient dispatch of renewable energy among interconnected microgrids. Thus, a negative ΔC in Fig. 6(b) represents the total saving in a multimicrogrid due to the interconnection. By adding ΔC to the annualized investment cost (C_{AnnPay}), we obtain the annual net economic saving (C_{Net}) of the interconnection. In Fig. 6(b), a negative C_{Net} shows that the alternative economically benefits the microgrid system.

We adopt the modified analysis in Fig. 6(b) because: it provides a clear comparison of reliability and economic benefits among different interconnection alternatives by assessing $\Delta C_{Interrupt}$ and it is able to consider the economic contribution of renewables in the planning algorithm by including ΔC_E . These two reasons correspond to the first and second benefits of interconnection presented in Section II.

In Fig. 6, the best option may not be the alternative with the lowest total cost. Rather, the final decision will consider several factors. Specifically, one may need to additionally assess each component in the total cost, evaluate different contributions on reliability improvement, and review other factors such as contractual and construction arrangements. Therefore, the next step in cost/worth analysis is to comprehensively examine the selected planning alternatives. Eleven alternatives for integrating wind farms were examined in [9] where the alternative with the second lowest total cost was considered as the best option, as it yields considerably higher reliability indices and also results in an integration arrangement that benefits the entire system.

In traditional reliability evaluation methods, the probabilistic index of loss-of-load-expectation (LOLE) was widely used for generating systems; expected-energy-not-served (EENS) and delivery-point-unavailability index (DPUI) were adopted for composite systems [20]; and loss-of-health-expectation (LOHE) was developed for small isolated systems with distributed generations [21].

Our proposed algorithm compares reliability performances of microgrids at different interconnection levels, e.g., microgrids *A* and *B* are compared with the interconnected system of *A* and *B*. So, in order to provide a reasonable and fair comparison, we adopt average indices, including ALOLE, AEENS, ADPUI, and ALOHE, where “A” stands for average

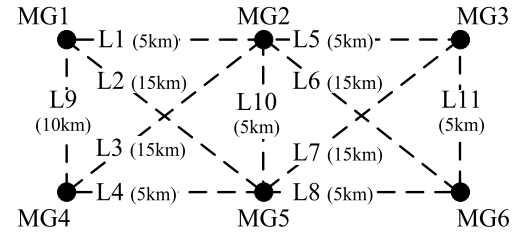


Fig. 7. Schematic for the proposed multimicrogrid interconnection.

in all cases. Thus, in the above example, we consider the following indices: LOLE of individual microgrids *A* and *B*, and ALOLE of the interconnected system of *A* and *B*.

The average indices consider that microgrids are essentially self-controlled composite generating entities while taking into account that multimicrogrids are small systems which are deployed close to each other at the distribution level. Here, two additional points need further clarifications. First, when studying a single microgrid, the average indices listed above are equivalent to standard reliability indices. Second, if assuming an equal number of customers in each microgrid, ALOLE and AEENS provide similar functions as distribution system indices, i.e., system-average-interruption-duration index (SAIDI) and average-energy-not-served (AENS), respectively, with a difference that ALOLE and AEENS concern microgrid-based operations while SAIDI and AENS are customer-based indices [20]. The average indices developed here will be further demonstrated in case studies in Section IV.

IV. CASE STUDIES FOR INTERCONNECTION PLANNING

This section provides the case studies using the proposed MCS-based methodology. The multimicrogrid system and the numerical results are presented in Sections IV-A and IV-B, respectively, and further discussions are given in Section IV-C. For the sake of discussion, let MG_i denote the i th microgrid, $MG_i \& j$ denote the interconnected system containing MG_i and MG_j , L denote interconnection line, and C denote MCS.

A. Study System

The studied multimicrogrid system including six individual microgrids is shown in Fig. 7, where dashed lines represent candidate interconnections. Historical data for wind speed, solar generation and load are taken in 10-min intervals for five years (July 2009 through June 2014) at IIT Microgrid are used in this paper.

The following parameter settings are presented in the case studies.

- 1) Table III(a) provides the peak-load and rated DER capacities for solar, wind and combined heat and power (CHP) units in each microgrid, where all numbers are in per unit with a 10.62 MW base (i.e., 2013 peak-load at IIT microgrid).
- 2) Table III(b) provides the economic data used in the case studies, where the data source is shown in the third column.

TABLE III

(a) MICROGRID GENERATION CAPACITIES AND PEAK-LOAD (P.U.).
 (b) RELIABILITY COST/WORTH PARAMETERS. (c) RELIABILITY LIMIT FOR EACH MICROGRID

(a)				
	Peak load (p.u.)	Rated DER capacity (p.u.)		
		Solar	Wind	CHP
MG1	1.0	1	1	-
MG2	1.0	-	-	-
MG3	1.0	0.5	-	1
MG4	0.9	1	1	-
MG5	0.9	-	-	-
MG6	0.9	0.5	-	1

(b)		
Economic data	Values adopted	Ref.
Length of lines (km)	Indicated in Fig. 7	
Cost of cable	1.24 M\$/km	[22]
Cost of power conversion devices	1.5 M\$/connection	[23]
Expected lifetime	40 years	[9]
Discount rate	8% per year	[9]
Value of lost load (VOLL)	8239 \$/MWh	[24]
Electricity price	0.16 \$/kWh	[25]

(c)		
Reliability indices	Limit	
AEENS (MWh/yr)	<200	
ALOLE (hr/yr)	<100	
ALOHE (hr/yr)	<140	
ADPUI (sys.hr)	<20	

- 3) Table III(c) gives the reliability constraints which microgrid planners seek to satisfy by an interconnection.
- 4) Multimicrogrids share generation reserves in island mode, while utilizing renewables in both grid-connected and island modes. The duration of islanded microgrid operation is assumed to be 3% of the total time. In grid-connected mode, the utility grid is assumed to be 100% reliable.
- 5) As we focus on interconnection planning, energy storage and load shedding schemes are not considered.

B. Numerical Results

1) *Data Processing*: The first step for applying the proposed planning methodology [see Fig. 5(a)] is the clustered data processing discussed in Section III-A. Fig. 8 shows the probabilistic wind generation distributions for 144 time periods (6×24 h), by applying SWM to historical wind speed data.

The three values (L , M , and H) in Fig. 8(a) represent the per unit wind energy production for three wind speed states (low, medium, and high) with the probabilities shown in Fig. 8(b).

Fig. 8 provides a probabilistic distribution model that takes into account the uncertainty and auto-correlation nature of wind. On the other hand, solar generation and load data are modeled using the averaging method discussed in Section III-A1. The processed load and generation data for each microgrid in Fig. 7 are scaled up/down in accordance with per unit peak-loads and rated DER in Table III(a). Based on processed data, the reliability of original system is evaluated with reliability indices matrix $\mathbf{R}_0$ shown in Table IV.

2) *Iterative Interconnection Planning*: In $\mathbf{R}_0$, MG1, MG3, and MG4 fail to satisfy the reliability limit

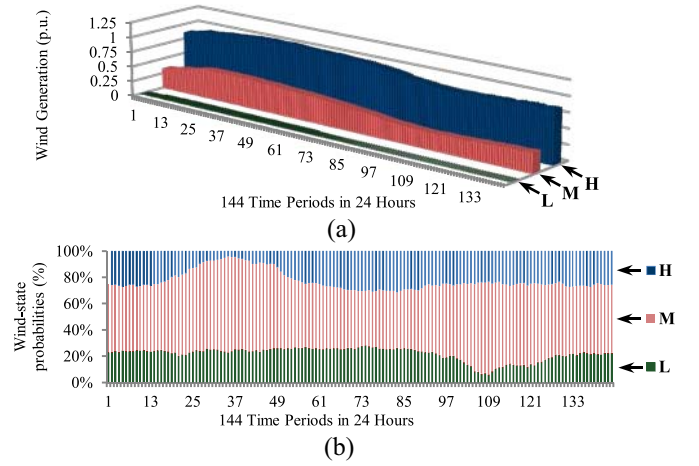


Fig. 8. SWM. (a) Wind generation (p.u.). (b) Wind-state probabilities.

TABLE IV
RELIABILITY INDICES MATRIX $\mathbf{R}_0$

Reliability indices	MG1	MG2	MG3	MG4	MG5	MG6
AEENS (MWh/yr)	347.6	182.2	128.1	256.8	123.6	63.1
ALOLE (hr/yr)	167.3	81.1	135.6	162.8	70.9	75.0
ALOHE (hr/yr)	169.7	87.8	176.0	165.7	74.5	133.0
ADPUI (sys.hr)	32.7	17.2	12.1	24.2	11.6	5.9

given in Table III(c). Therefore, we apply the iterative procedure devised in Fig. 5(a) for the interconnection planning. The results in each iteration are presented below.

Iteration 1: Since MG1 has the worst reliability performance among MG1, MG3, and MG4, the objective in iteration 1 is to seek an interconnection solution that will improve the MG1 reliability.

a) *Application of the MCS method*: Probabilistic microgrid status combinations at each time period are identified following the discussion in Section III-B, where the numbers of time periods t with the stored status ($t \in T$) are: 38 616, 36 943, and 35 448 for wind states L , M , and H , respectively. The proposed probabilistic MCS method is applied to the stored status and the results are listed in Table V(a). In this table, each row represents an identified MCS; each code in the second column denotes a microgrid status combination (e.g., “010010” indicates that MG2 and MG5 are surplus while the others are deficient) and the fourth column gives the occurrence probability of each MCS which is calculated by assessing the wind-state probabilities in each time period.

Specifically, the top ranked MCSs in Table V(a) presents a notion that the established interconnection for delivering energy from {MG2, MG5} to {MG1, MG3, MG4, MG6} can potentially improve the reliability and the economic performance of the multimicrogrid system. Specifically, MCSs C1–C4 obtain high probabilities, which is due to the complementarity between solar and wind energy in MG1, MG2, MG4, and MG5. For example, C1 represents the situations when wind is high and solar is low (mostly at night) such that the surplus wind energy provided by MG2 and MG5 could help serve the loads in MG1 and MG4; while C4 represents

TABLE V
(a) RESULTS OF PROBABILISTIC MCS METHOD (ITERATION 1).
(b) RANK OF LINES WITH PROBABILITIES (ITERATION 1)

(a)				(b)			
MCS	MG status	Prob.	Lines in MCS	MCS	MG status	Prob.	Lines in MCS
C1	010010	0.1738	L1-8	C7	000010	0.0155	L2, L4, L7-8, L10
C2	010011	0.1611	L1-4, L5, L7, L11	C8	101111	0.0133	L1, L3, L5-6, L10
C3	011011	0.1180	L1-4	C9	110111	0.0111	L5, L7, L11
C4	101101	0.1014	L1-8	C10	100100	0.0104	L1-4
C5	000001	0.0429	L6, L8, L11	C11	100101	0.0084	L1-4, L6, L8, L11
C6	001001	0.0340	L5-8	C12	110110	0.0052	L5-8

Line	Prob.	Line	Prob.
L2	0.5886	L8	0.3811
L4	0.5886	L6	0.3790
L1	0.5865	L11	0.2234
L3	0.5865	L10	0.0288
L7	0.5021	L9	0.0000
L5	0.4999		

TABLE VI
RELIABILITY COST/WORTH ANALYSIS (ITERATION 1)

		L2 (MG1-MG5)	L1 (MG1-MG2)
Reliability Indices	AEENS (MWh/yr)	105.9	118.6
	ALOLE (hr/yr)	40.8	41.7
	ALOHE (hr/yr)	47.2	46.7
	ADPUI (sys.hr)	10.0	11.2
	$\Delta C_{Interrupt}$ (M\$/yr)	-2.14	-2.19
Cost/worth analysis	ΔC_E (M\$/yr)	-1.78	-1.80
	C_{AnnPay} (M\$/yr)	0.84	0.32
	C_{Net} (M\$/yr)	-3.07	-3.67

the counter situation (mostly during daytime). This case demonstrates the merit of microgrid interconnection.

Based on the discussion in Section III-B, the MCS probability in Table V(a) is assigned to each line included in that MCS; then candidate lines recommended by their accrued possibilities are listed in Table V(b). In this table, lines L1–L4 are top ranked due to the high probabilities of C1–C4, which are ultimately derived based on various penetration levels of solar and wind generations in MG1, MG2, MG4, and MG5.

b) *Reliability cost/worth analysis*: Since the purpose of this iteration is to improve the reliability of MG1, only those lines that are connected to MG1 (i.e., L1, L2, and L9) are reviewed. In Table V(b), L1 and L2 are top-ranked while L9 obtains nearly a zero probability as L9 connects MG1 to MG4 but MG4 is also unreliable. The results in Table V(b) verify that the proposed planning method is able to efficiently rank the candidate lines based on their potential reliability contribution.

Since L1 and L2 have close probability values in Table V(b), the reliability cost/worth analysis discussed in Section III-D is applied to these two alternatives with results shown in Table VI. Here, the developed indices are further illustrated by using L2 as an example. That is, indices corresponding to L2 reflect the average reliability performance of MG1 and MG5 by considering the two microgrids as a single system since they are connected by L2. In Table VI, both L1 and L2 contribute considerable economic savings as indicated by $\Delta C_{Interrupt}$ and ΔC_E , both of which can be

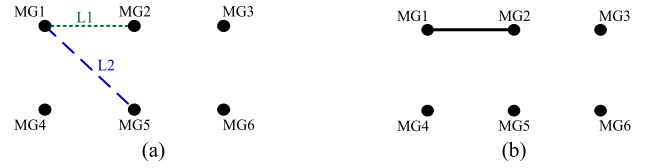


Fig. 9. Network topology in iteration 1. (a) Two alternatives. (b) Solution.

TABLE VII
RELIABILITY INDICES MATRIX $\mathbf{R}_1$

Reliability indices	MG1&2	MG3	MG4	MG5	MG6
AEENS (MWh/yr)	118.6	128.1	256.8	123.6	63.1
ALOLE (hr/yr)	41.7	135.6	162.8	70.9	75.0
ALOHE (hr/yr)	46.7	176.0	165.7	74.5	133.0
ADPUI (sys.hr)	11.2	12.1	24.2	11.6	5.9

TABLE VIII
(a) RESULTS OF PROBABILISTIC MCS METHOD (ITERATION 2).
(b) RANKING OF LINES WITH PROBABILITIES (ITERATION 2)

(a)				(b)			
MCS	MG status	Prob.	Lines in MCS	MCS	MG status	Prob.	Lines in MCS
C1	00010	0.1218	L2, L4, L7-8, L10	C7	00001	0.0429	L6, L8, L11
C2	11011	0.1166	L3, L4, L9	C8	01001	0.0340	L5-L8
C3	10011	0.1067	L2-7, L9, L10	C9	10111	0.0167	L5, L7, L11
C4	11101	0.0947	L2, L4, L7-8, L10	C10	00100	0.0093	L3, L4, L9
C5	10010	0.0674	L3-9	C11	10110	0.0082	L5-L8
C6	00011	0.0545	L2, L4, L6-7, L10-11	C12	01101	0.0082	L3-L8, L9

Line	Prob.	Line	Prob.
L4	0.5791	L6	0.3219
L7	0.5124	L3	0.3081
L2	0.3777	L9	0.3081
L10	0.3777	L5	0.2413
L8	0.3773	L11	0.2207

attributed to the solar-wind complementarity in interconnected microgrids. Here, it is noted that from the reliability point of view, indices corresponding to L2 are slightly better than those of L1, which is due to the MG5s slightly lower peak-load; however, L1 contributes more cost-saving due to interruptions and more efficient dispatch of renewables while it also lowers the project investment. Therefore, L1 is comprehensively more advantageous and thus considered as the optimal option to improve the reliability of MG1 in this iteration.

c) *Updating network topology and reliability indices*: The updated network topology with L1 constructed is shown in Fig. 9(b). Updated reliability indices matrix ($\mathbf{R}_1$) is given in Table VII.

Iteration 2: According to $\mathbf{R}_1$, MG3 and MG4 still fail to satisfy the reliability limit given in Table III(c), thereby requiring more iterations. This iteration will aim at the MG4 status as it has the worst reliability indices in $\mathbf{R}_1$.

a) *Application of the MCS method*: Based on the updated topology in Fig. 9(b), microgrid status combinations are reidentified, where the numbers of time periods t with the stored status ($t \in T$) are: 38 616, 35 148, and 35 118 for wind-states L , M , and H , respectively. The MCS method is applied with results shown in Table VIII(a) in which the first digit of the status code denotes the interconnected MG1&2.

TABLE IX
RELIABILITY COST/WORTH ANALYSIS (ITERATION 2)

		L4 (MG4-MG5)	L3 / L9 (MG4-MG1&2)
Reliability Indices	AEENS (MWh/yr)	83.3	144.8
	ALOLE (hr/yr)	35.1	36.3
	ALOHE (hr/yr)	41.7	41.7
	ADPUI (sys.hr)	7.8	13.6
Cost/worth analysis	$\Delta C_{Interrupt}$ (M\$/yr)	-1.76	-0.71
	ΔC_E (M\$/yr)	-1.76	-0.14
	C_{AnnPay} (M\$/yr)	0.32	0.56 / 0.39
	C_{Net} (M\$/yr)	-3.20	-0.29 / -0.46

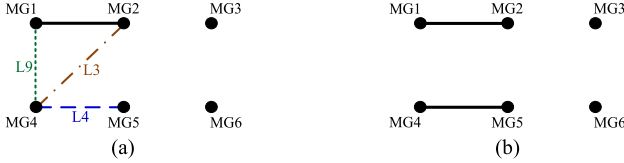


Fig. 10. Network topology in iteration 2. (a) Two alternatives. (b) Solution.

TABLE X
RELIABILITY INDICES MATRIX $\mathbf{R}_2$

Reliability Indices	MG1&2	MG3	MG4&5	MG6
AEENS (MWh/yr)	118.6	128.1	83.3	63.1
ALOLE (hr/yr)	41.7	135.6	35.1	75.0
ALOHE (hr/yr)	46.7	176.0	41.7	133.0
ADPUI (sys.hr)	11.2	12.1	7.8	5.9

In this table, the top-ranked C1–C3 imply that MG4 may seek assistance from (MG1&2), MG5, or MG6. Here, MG5 draws more attention as the only surplus microgrid through C1–C3.

Using the assigned probabilities, the interconnection lines are ranked in Table VIII(b) in which L3, L4, and L9 are reviewed as they are connected to MG4. Here, L4 is top-ranked because it can efficiently realize the solar-wind complementarity between MG4 and MG5, which also agrees with our earlier prediction concerning MG5. L3 and L9 on the other hand rank much lower. Therefore, in this iteration L4 can be efficiently selected as the optimal option, i.e., this iteration is completed and reliability cost/worth analysis is dispensable.

b) Reliability cost/worth analysis: The reliability cost/worth analysis in this iteration is not imperative for decision making; however, it is used to verify our earlier conclusion regarding L4. In Table IX, L4 offers a much higher reliability while at the same time it contributes to savings. Therefore, the earlier conclusion is verified and the results have demonstrated the accuracy and higher efficiency of the proposed methodology.

c) Updating the network topology and reliability indices: Fig. 10(b) shows the updated topology and $\mathbf{R}_2$ is given in Table X.

Iteration 3: Since MG3 in $\mathbf{R}_2$ still does not satisfy the reliability limit, an additional iteration is required for MG3.

a) Application of the MCS method: Based on the updated topology in Fig. 10(b), microgrids status combinations

TABLE XI
(a) RESULTS OF PROBABILISTIC MCS METHOD (ITERATION 3).
(b) RANK OF LINES WITH PROBABILITIES (ITERATION 3)

(a)				(b)			
MCS	MG status	Prob.	Lines in MCS	MCS	MG status	Prob.	Lines in MCS
C1	1011	0.1282	L5, L7, L11	C5	0101	0.0381	L5-L8
C2	1010	0.0792	L5-L8	C6	0010	0.0292	L2-3, L7-10
C3	0001	0.0518	L6, L8, L11	C7	0111	0.0056	L2-3, L5-6, L9-10
C4	0011	0.0505	L2-3, L6-7, L9-11				

Line	Prob.	Line	Prob.
L7	0.3253	L2	0.0853
L5	0.2511	L3	0.0853
L6	0.2253	L9	0.0853
L8	0.1984	L10	0.0853
L11	0.1801		

TABLE XII
RELIABILITY COST/WORTH ANALYSIS (ITERATION 3)

		L7: (MG3-MG4&5)	L5: (MG3-MG1&2)
Reliability Indices	AEENS (MWh/yr)	83.8	119.9
	ALOLE (hr/yr)	27.1	32.9
	ALOHE (hr/yr)	33.2	38.9
	ADPUI (sys.hr)	7.9	11.3
Cost/worth analysis	$\Delta C_{Interrupt}$ (M\$/yr)	-0.36	-0.26
	ΔC_E (M\$/yr)	-0.54	-0.40
	C_{AnnPay} (M\$/yr)	0.22	0.22
	C_{Net} (M\$/yr)	-0.68	-0.44



Fig. 11. Network topology in iteration 3. (a) Three alternatives. (b) Solution.

are reidentified, where the number of time periods t with the stored status ($t \in T$) are: 22 038/16 229/26 759 (L/M/H wind-states). Then the MCS method is applied with results shown in Table XI(a). The results imply that MG3 may seek assistance from MG1&2, MG4&5, or MG6, where more attention is paid to MG1&2 and MG4&5 as they both have a surplus status through C1–C2. With possibilities assigned, lines are ranked in Table XI(b), where L5, L7, and L11 are reviewed as they are connected to MG3. Here, L11, which is connected to MG6, gains a much lower probability than L5 and L7.

b) Reliability cost/worth analysis: Due to L11's low probability, the reliability cost/worth analysis is conducted on L7 and L5 with results shown in Table XII. As compared with L5, L7 yields close economic savings but offers much better reliability indices; so L7 is selected as the optimal option.

c) Updating network topology and reliability indices: Fig. 11(b) shows the updated topology and $\mathbf{R}_3$ is given in Table XIII. Since each microgrid in $\mathbf{R}_3$ has satisfied the given

TABLE XIII
RELIABILITY INDICES MATRIX $\mathbf{R}_3$

Reliability Indices	MG1&2	MG3&4&5	MG6
AEENS (MWh/yr)	118.6	83.8	63.1
ALOLE (hr/yr)	41.7	27.1	75.0
ALOHE (hr/yr)	46.7	33.2	133.0
ADPUI (sys.hr)	11.2	7.9	5.9

reliability limits, the iteration is terminated and the topology shown in Fig. 11(b) is the final solution for the planning problem.

C. Further Discussion of the Results

First, the final topology shown in Fig. 11(b) suggests the clustering of microgrids into three groups: 1) MG1&2; 2) MG3&4&5; and 3) MG6. In these three groups, MG1&2 realizes the complementarity between solar and wind generations; MG3&4&5 also provides supreme features while including a microgrid with larger CHP capacity thus building a multimicrogrid system with hybrid generation; MG6 as a CHP dominated microgrid with lower peak-load has already satisfied the reliability constraint therefore does not require any interconnection. On the other hand, it is expected that if the multimicrogrid system has loads with very high Value of Lost Load (e.g., data centers or hospitals), the solution may suggest interconnecting more individual microgrids.

Second, significant improvements in reliability are observed by comparing $\mathbf{R}_3$ to $\mathbf{R}_0$. Such improvements are mainly because the DER integration can mitigate the variability of renewables and complement solar with wind energy. This also verifies the benefits of microgrid interconnections.

Third, for the sake of discussion, a six-microgrid system is used in our case studies. The proposed methodology can be readily applied to other multimicrogrid systems. It is expected that a higher number of participating microgrids will result in a higher efficiency of the proposed methodology, as we only need to focus on top-ranked interconnection lines. For example, in a large multimicrogrid system where a single microgrid has N candidate interconnection lines, only the top-ranked two or three lines, with similar probabilities will be considered for reliability cost/worth analyses to determine the optimal interconnection planning solution.

Finally, although each iteration in our case study addressed a single microgrid, there is a possibility that multimicrogrids can change from reliability-unsatisfied to reliability-satisfied operations in the same iteration.

V. CONCLUSION

This paper proposed the application of a probabilistic MCS-based iterative methodology for the optimal planning of multimicrogrid interconnection. Various concerns including economics, reliability, and generation variability are addressed in the optimal interconnection planning. The proposed methodology adopts various techniques pertaining to the power system planning including SWM, microgrid status analysis, probabilistic MCS method, and reliability cost/worth analyses.

Suitable indices are also developed for the reliability evaluation of interconnected microgrids.

The proposed methodology is applied to a six-microgrid system operating in island mode. The numerical results suggest the installation of three interconnection lines to ensure each microgrid will satisfy the given reliability constraints while satisfying the limited capital cost. The numerical results have successfully demonstrated that the proposed interconnection planning methodology is able to accurately rank interconnection alternatives by indicating their potential reliability contributions and efficiently suggest an optimal interconnection planning option. The numerical results also verified that the proposed methodology is adaptive and can be implemented easily for practical microgrid applications.

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